

재료상변태(공) (MSE402-01)

학기	2026년 1학기
담당교수	김진수
이수구분	교선평수
학점	3.0
강의시간	수10:30-12:00(E7 - 223), 금10:30-12:00(E7 - 223)
강의실	E6 315 (교수실)
소속	기초학부

교과개요

This course aims to learn the basic concepts of phase transformation in metals and alloys and understand the effects of microstructural changes on the mechanical properties and characteristics of materials. This course will cover diffusion, nucleation, growth, and transformation behavior based on thermodynamic and kinetic principles.

교과목표

1. Understand the basic concepts of phase transformation and thermodynamic and kinetic principles. 2. Analyze the microstructural changes that occur during processes. 3. Acquire basic theories such as solid-state phase transformation, diffusion-based phase transformation, and diffusionless transformation. 4. Learn cases so that can be applied to actual materials science related circumstances.

주차별 강의계획

1주차:Thermodynamics and Phase Diagrams (1) - Single Component Systems, Binary Solutions
2주차:Thermodynamics and Phase Diagrams (2) - Binary Phase Diagrams
3주차:Diffusion (1) - Interstitial Diffusion
4주차:Diffusion (2) - Substitutional Diffusion
5주차:Diffusion (3) - High-Diffusivity Paths, Diffusion in Multiphase Binary Systems
6주차:Crystal Interfaces and Microstructure (1) - Boundaries in Single-Phase Solids
7주차:Crystal Interfaces and Microstructure (2) - Interphase Interfaces in Solids
8주차:Midterm Exam
9주차:Solidification (1) - Nucleation in Pure Metals, Growth of a Pure Solid
10주차:Solidification (2) - Alloy Solidification, Solidification of Ingots and Castings
11주차:Diffusional Transformations in Solids (1) - Heterogeneous Nucleation, Precipitate Growth
12주차:Diffusional Transformations in Solids (2) - Precipitation in Age-Hardening Alloys
13주차:Diffusional Transformations in Solids (3) - Eutectoid Transformations
14주차:Diffusionless Transformations (1) - Characteristics of Diffusionless Transformations, Martensite
Crystallography
15주차:Diffusionless Transformations (2) - Theories of Martensite Nucleation, Martensite Growth
16주차:Final Exam

학습윤리

Unauthoritarian assistance is not allowed on all assignments and exams. Misrepresentation of your identity is not also permitted.

교과정책(방침)

The course will consist mainly of theoretical lectures, but will include some practical and case studies. Students are required to submit assignments on the given date, and late submissions will result in a deduction of points. Additional materials (papers, experimental results, etc.) will be provided when necessary. Midterm and final exams are not open-book.